Software-Hardware Contract for Side Channel Defenses

Mengjia Yan

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Attack Examples

Example #1: termination time vulnerability

```
def check_password(input):

  size = len(password);

  for i in range(0,size):
    if (input [i] != password[i]):
      return ("error");


  return (“success”);
```

Example #2: RSA cache vulnerability

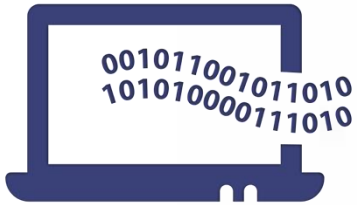
```
for i = n-1 to 0 do
  r = sqr(r)
  r  = r mod n
  if e_i == 1 then
    r = mul(r, b)
    r  = r mod n
  end
end
```

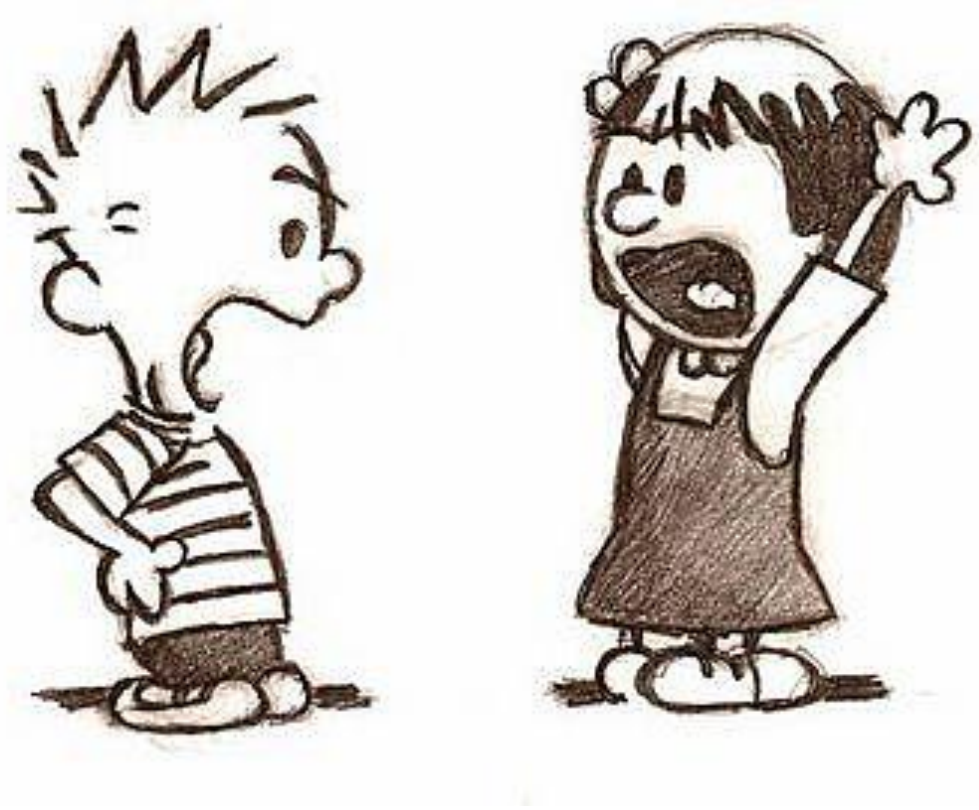
Example #3: Meltdown

```
……
Ld1: uint8_t secret = *kernel_address;
Ld2: unit8_t dummy = probe_array[secret*64];
```

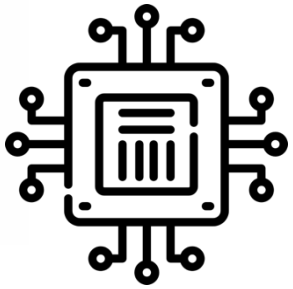
Who to blame? Who to fix the problem?

Software Developers

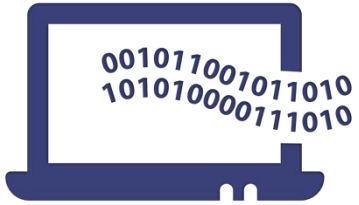




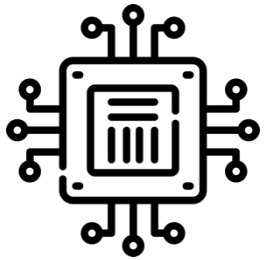
Hardware Designers



These Attacks Break SW-HW Contract



Software



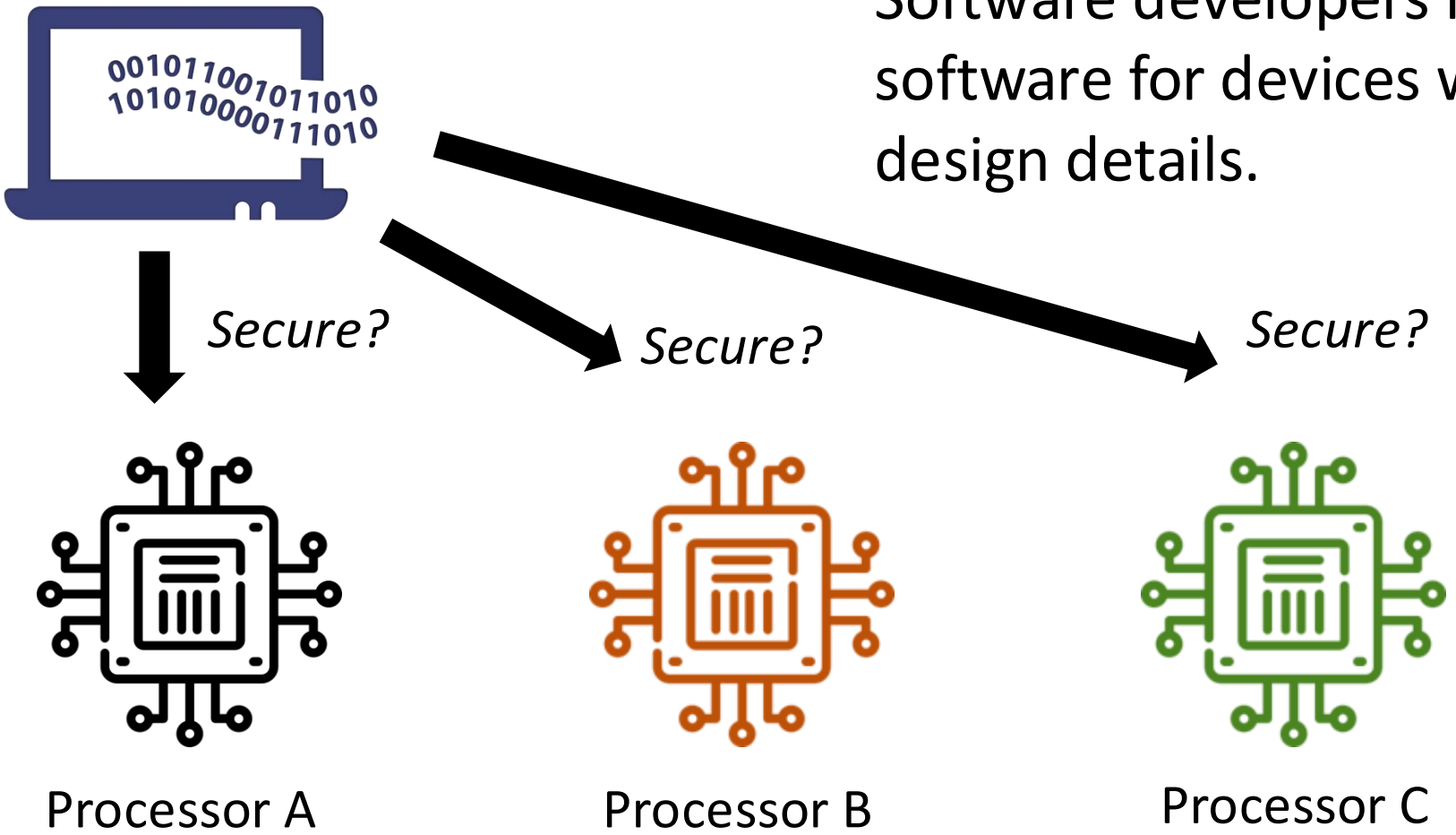
Hardware

Instruction Set Architecture (ISA)

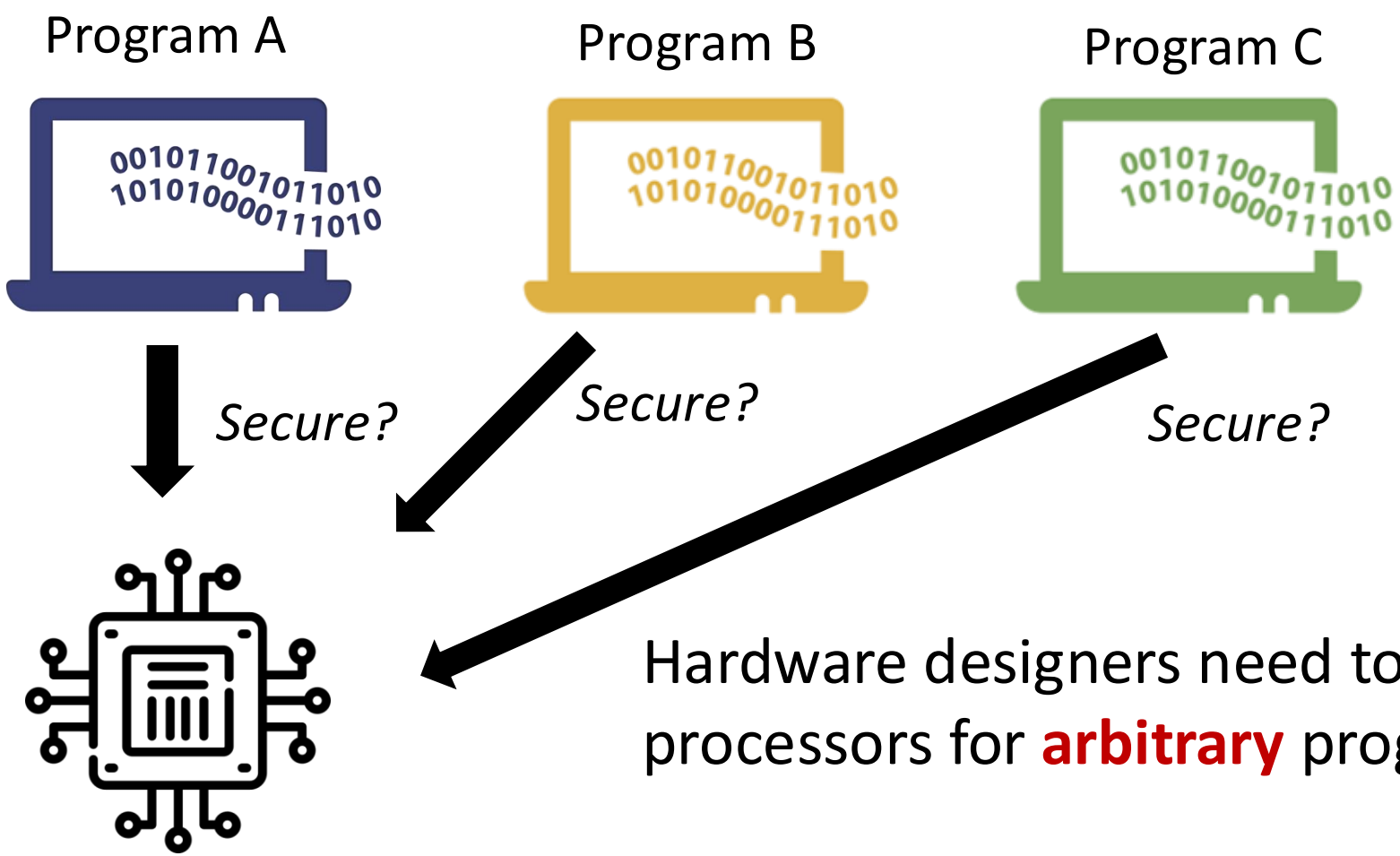
The contract for functional correctness.

Software Developer's Problem

Software developers need to write software for devices with **unknown** design details.



Hardware Designer's Problem



Hardware designers need to design processors for **arbitrary** programs.

Example: Termination Time Vulnerability

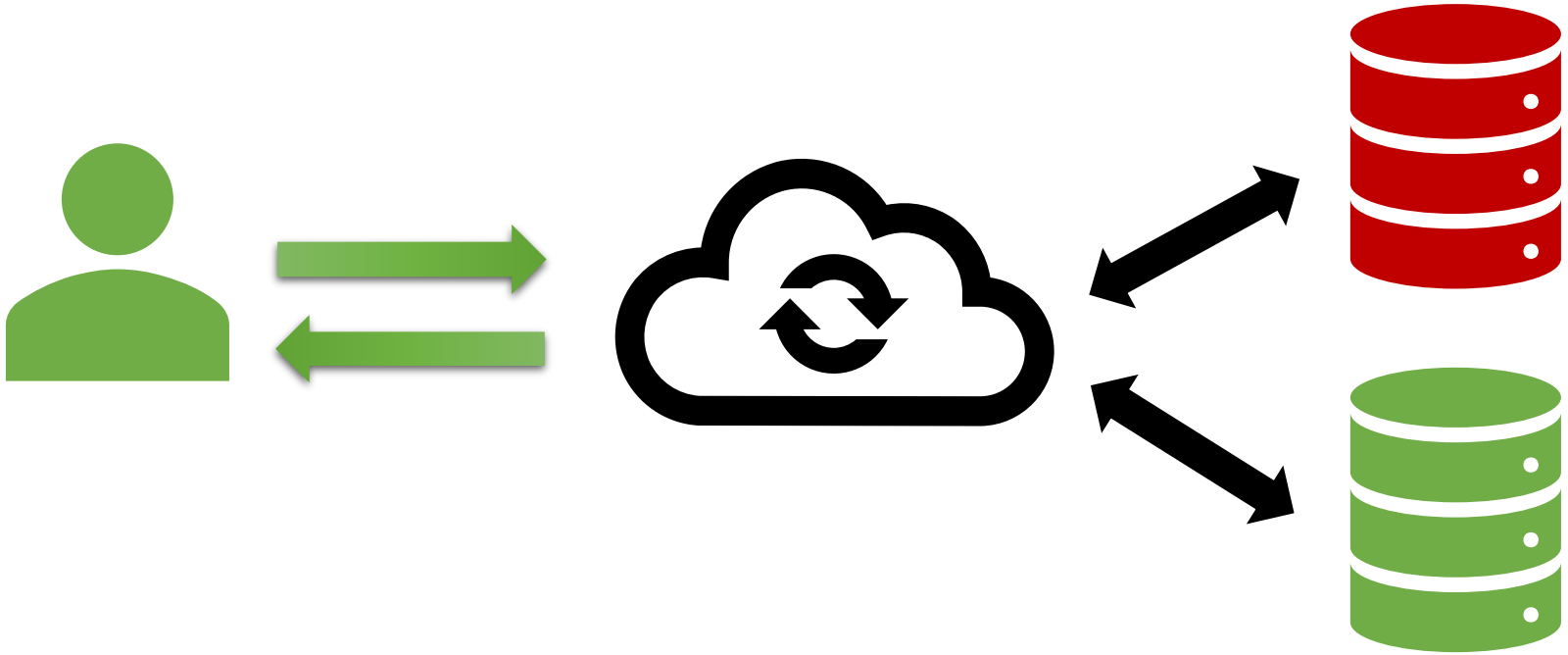
- How to fix it?

```
def check_password(input):

  for i in range(0,128):
    if (input [i] != password[i]):
      return ("error");


  return (“success”);
```

Make the computation time **independent** from the secret

What do we mean by “independent”? Let’s be a bit more **rigorous**.

Non-Interference Example



- Intuitively: not affecting
- Any sequence of **low** inputs will produce the same **low** outputs, regardless of what the **high** level inputs are.
- Example: a password box

Non-Interference: A Formal Definition

- The definition of noninterference for a deterministic program P

$$\forall\ \mathrm{M1}, \mathrm{M2}, \mathrm{P}$$

$$\mathrm{M1_L} = \mathrm{M2_L} \quad \wedge \quad (\mathrm{M1}, \mathrm{P}\,) \rightarrow^{*} \mathrm{M1}' \quad \wedge \quad (\mathrm{M2}, \mathrm{P}\,) \rightarrow^{*} \mathrm{M2}'$$

$$\Longrightarrow \quad \mathrm{M1}'_{\mathrm{L}} = \mathrm{M2_L}'$$

Non-Interference for Side Channels

- The definition of noninterference for a deterministic program P

$$\forall\, \mathrm{M1}, \mathrm{M2}, \mathrm{P}$$

$$\mathrm{M1_L} = \mathrm{M2_L} \quad \wedge \quad (\mathrm{M1}, \mathrm{P}) \xrightarrow{\mathrm{O1}}{}^{*}\, \mathrm{M1}' \quad \wedge \quad (\mathrm{M2}, \mathrm{P}) \xrightarrow{\mathrm{O2}}{}^{*}\, \mathrm{M2}'$$

$$\Longrightarrow \quad \mathrm{O1} = \mathrm{O2}$$

What should be included in the observation trace (O)?

Instruction completion time

Operands of non-constant time instructions, such as addresses issued to the memory systems (for both data and instruction)

Understand the Property

$$\forall\, M1, M2, P$$

$$M1_L = M2_L \;\wedge\; (M1, P) \xrightarrow{O1}{}^{*} M1' \;\wedge\; (M2, P) \xrightarrow{O2}{}^{*} M2'$$

$$\Longrightarrow \quad O1 = O2$$

```
def check_password(input):

  for i in range(0,128):
    if (input [i] == password[i]):
      return ("error");


  return (“success”);
```

Consider input as part of M

- What is M_L ?
- What is M_H ?
- What is O ?

Constant-Time Programming

Think about whether the statement below is true or false.

- For any public inputs, secret values, and machines, a program always takes the same amount of time to execute.
- For any public inputs, secret values, a program always takes the same amount of time when executing on the same machine.
- For any secret values, a program always takes the same amount of time for the same public input when executing on the same machine.
- For any secret values, a program always takes the same amount of time for the same input when executing on the same machine, and this holds for arbitrary public inputs.

Data-oblivious/Constant-time programming

- How to deal with conditional branches/jumps?
- How to deal with memory accesses?
- How to deal with arithmetic operations: division, shift/rotation, multiplication?

Your Code

Compiler

Hardware

For details on real-world constant-time crypto, check this out: https://www.bearssl.org/constanttime.html

```
def check_password(input):

  for i in range(0,128):
    if (input [i] != password[i]):
      return ("error");


  return (“success”);
```

```
def check_password(input):

  dontmatch = false;
  for i in range(0,128):
    if (input [i] != password[i]):
      dontmatch = true;


return dontmatch;
```

```
def check_password(input):

  dontmatch = false;
  for i in range(0,128):
    if (input [i] != password[i]):
      dontmatch = true;

return dontmatch;
```

```
def check_password(input):

  dontmatch = false;
  for i in range(0,128):
    dontmatch |= (input [i] != password[i])

  return dontmatch;
```

Real-world Crypto Code

From libsodium cryptographic library:

```
for (i = 0; i < n; i++)
    d |= x[i] ^ y[i];
return (1 & ((d - 1) >> 8)) - 1;
```

What do we **assume** about the hardware here?

Compare two buffers x and y, if match, return 0, otherwise, return -1.

Examples from Cauligi et al. FaCT: A DSL for Timing-Sensitive Computation. PLDI'19

Another Example

From the “donna” Curve25519 implementation

```
for (i = 0; i < 5; ++i)
{
    if (swap) {
        tmp = a[i];
        a[i] = b[i];
        b[i] = tmp;
    }
}
```

```
for (i = 0; i < 5; ++i) {
    const limb x = swap & (a[i] ^ b[i]);
    a[i] ^= x;
    b[i] ^= x;
}
```

swap is a mask, either 0 or 0xFFFFFFFF

Eliminate Secret-dependent Branches

- Be a master of bitmask operations

- An instruction: `cmov_`
  - Check the state of one or more of the status flags in the EFLAGS register (`cmovz`: moves when ZF=1)
  - Perform a move operation if the flags are in a specified state
  - Otherwise, a move is not performed (as if a NOP) and execution continues with the instruction following the `cmov` instruction

Conditional Branches

- Original program

```
if (secret) x = e
```

- Use bitmask

```
x = (-secret & e) | (secret - 1) & x
```

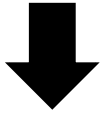
- Use cmov

```
test secret, secret // set ZF=1 if zero
cmovz r2, r1 // r2 has x, r1 has e
```

What do we **assume** about the hardware here?
(Hint: there are two.)

More Conditional Branches

```
if (secret)
  res = f1();
else
  res = f2();
```



```
r1 ← f1();
r2 ← f2();
mov r3, r1
test secret, secret
cmovz r3, r2
// res in r3
```

Potential problems:

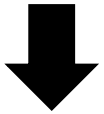
- What if we have nested branches?
- What if when `secret==0`, `f1` is not executable, e.g., causing page fault or divide by zero?
- What if `f1` or `f2` needs to write to memory, perform IO, make system calls?

Data-oblivious/Constant-time programming

- How to deal with conditional branches/jumps? ✅
- How to deal with memory accesses?
- How to deal with arithmetic operations: division, shift/rotation, multiplication?

Memory Accesses

```
a = buffer[secret]
```



```
for (i=0; i<size; i++)
{
    tmp = buffer[i];
    xor secret, I //set ZF
    cmovz a, tmp
}
```

- Performance overhead.
- Techniques such as ORAM can reduce the overhead when the buffer is large

An Optimization

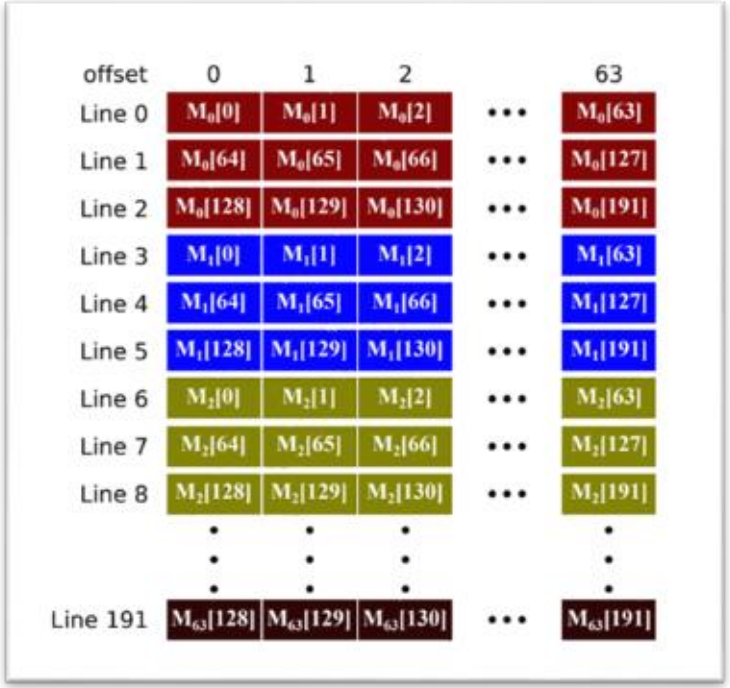
- Proposal: reduce the redundant accesses by only accessing one byte in each cache line.

```
for (i=0; i<size; i++)
{
    tmp = buffer[i];
    xor secret, i
    cmovz a, tmp
}
```

```
offset = secret % 64;
for (i=0; i<size; i+=64)
{
    index = i + offset;
    tmp = buffer[index];
    xor secret, index
    cmovz a, tmp
}
```

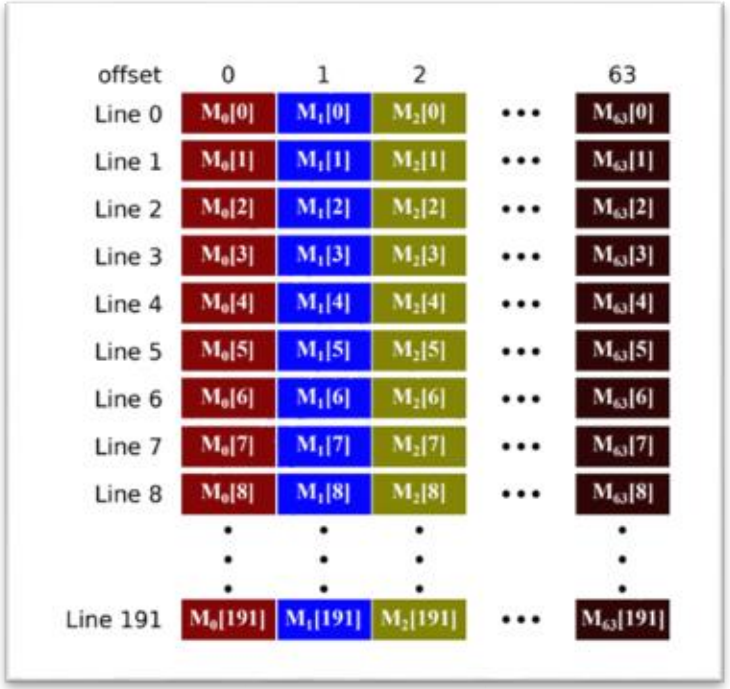
What do we assume about the hardware here?

OpenSSL Patches Against Timing Channel



Conventional Layout

Vulnerable to traditional cache attacks ?



Scatter Layout

to mitigate cache attacks

Vulnerable to L1 bank conflict attacks ?

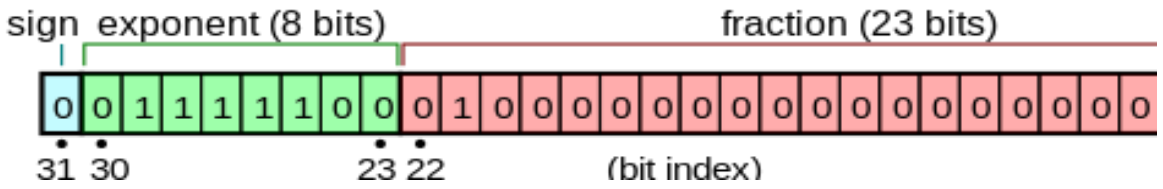
Yarom et al. CacheBleed: A Timing Attack on OpenSSL Constant Time RSA.
https://faculty.cc.gatech.edu/~genkin/cachebleed/index.html

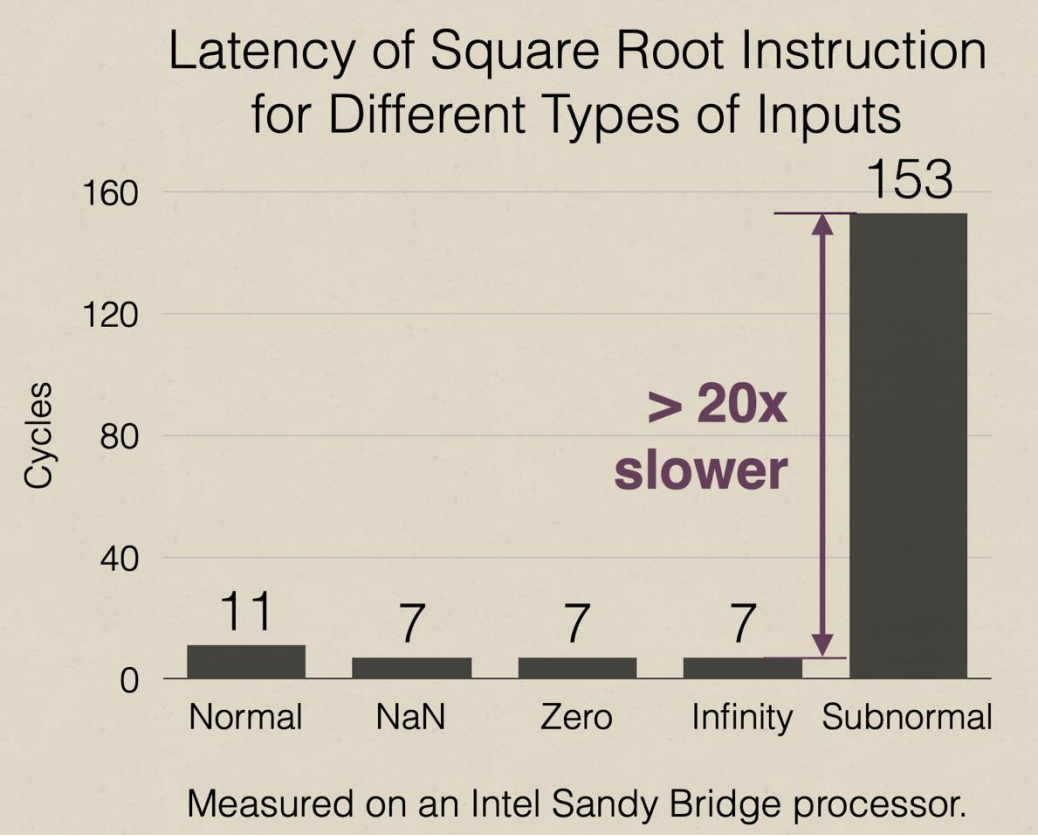
Data-oblivious/Constant-time programming

- How to deal with conditional branches/jumps? ✅
- How to deal with memory accesses? ✅
- How to deal with arithmetic operations: division, shift/rotation, multiplication?

Arithmetic Operations

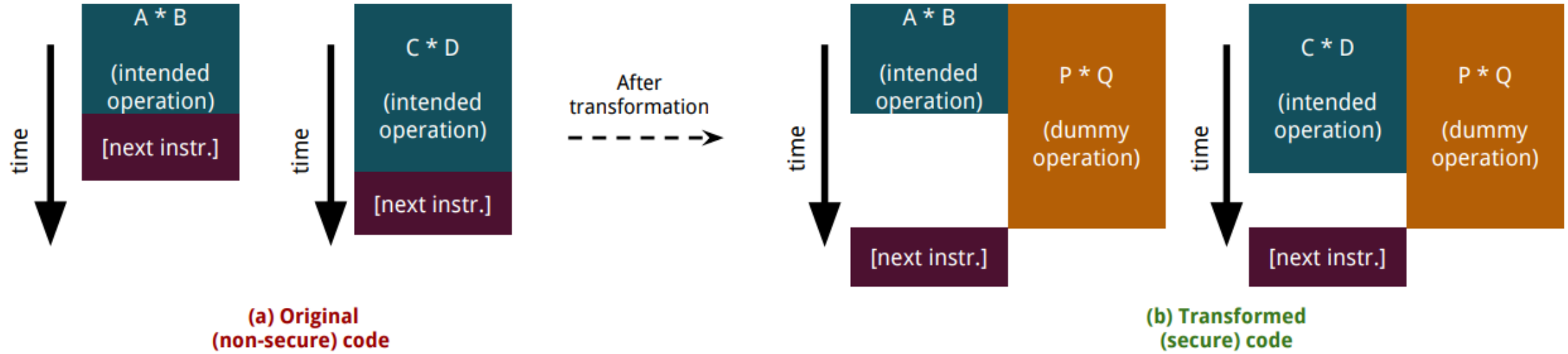
Subnormal floating point numbers





Andrysco et al; On Subnormal Floating Point and Abnormal Timing; S&P'15

The Problem and A Solution

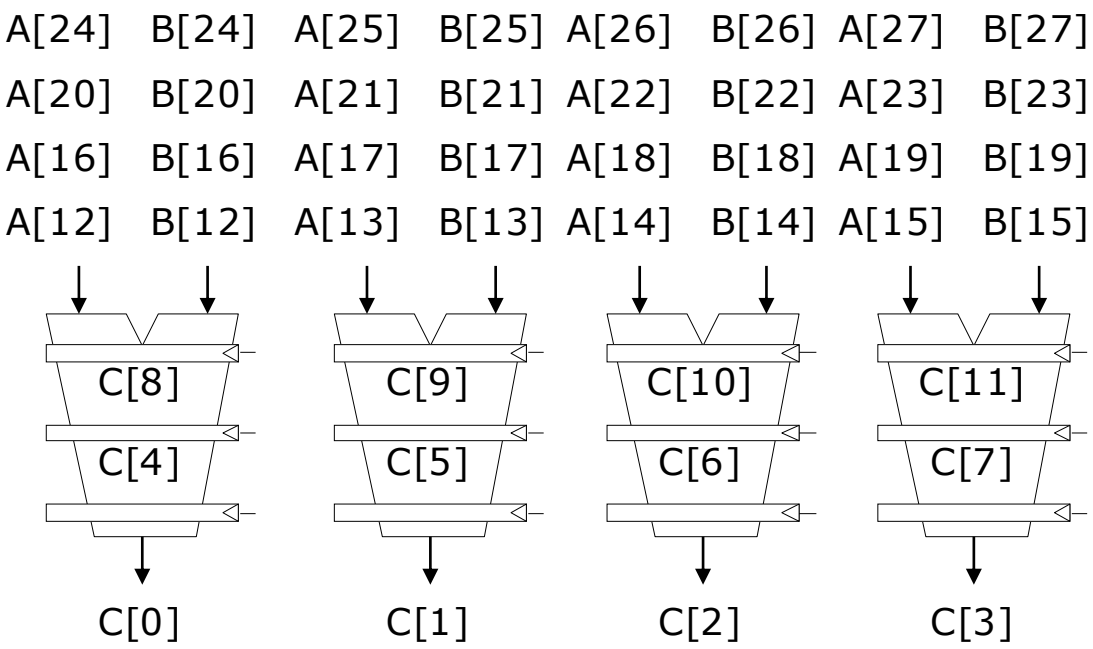


Rane et al. Secure, Precise, and Fast Floating-Point Operations on x86 Processors. USENIX'16

Single Instruction Multiple Data (SIMD)

```
# Vector code
LI VLR, 64 //length
LV V1, R1  // vec 1
LV V2, R2  // vec 2
ADDV.D V3, V1, V2
SV V3, R3
```

Example: 4 pipelined functional units

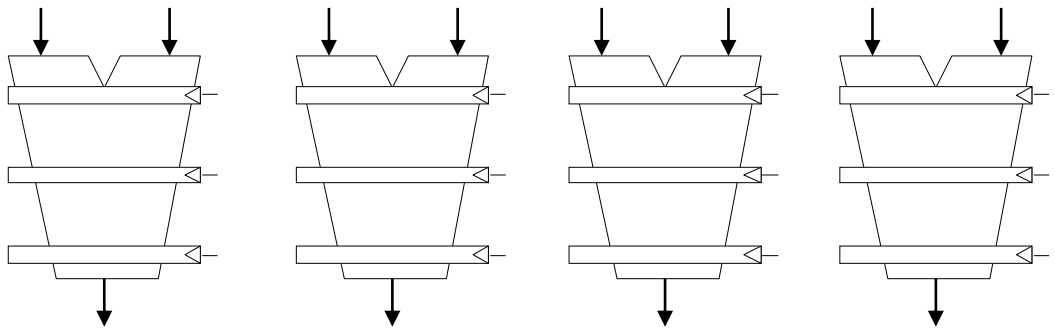


Make Floating-Point Constant Time

What do we assume about the hardware here?

Parameters for the actual computation

Selected subnormal numbers



Hardware Assumption:

1. The selected subnormal number takes the maximum length
2. SIMD returns only if the slowest lane finishes

How shall we proceed?

- The key problem:
  - No **explicitly** SW-HW contract for timing
  - SW developers derive hardware assumptions from *existing attacks* and impose **implicit** assumptions on the hardware.

- Some incoming efforts:
  - ARM Data Independent Timing (DIT)
  - Intel Data Operand Independent Timing (DOIT)

ARM DIT: https://developer.arm.com/documentation/ddi0601/2020-12/AArch64-Registers/DIT--Data-Independent-Timing
Intel DOIT: https://www.intel.com/content/www/us/en/developer/articles/technical/software-security-guidance/best-practices/data-operand-independent-timing-isa-guidance.html

So far, we have not discussed how to deal with speculation…

What’s Next?

- Mitigations of transient execution attacks
  - By Yuheng Yang
  - Fancy interactive simulator to visualize transient execution
- Physical attacks
  - By Joseph Ravichandran
  - Three in-class real-time demos of physical attacks
- Embedded system attack CTF (recitation)
  - Another CTF, prize for winners